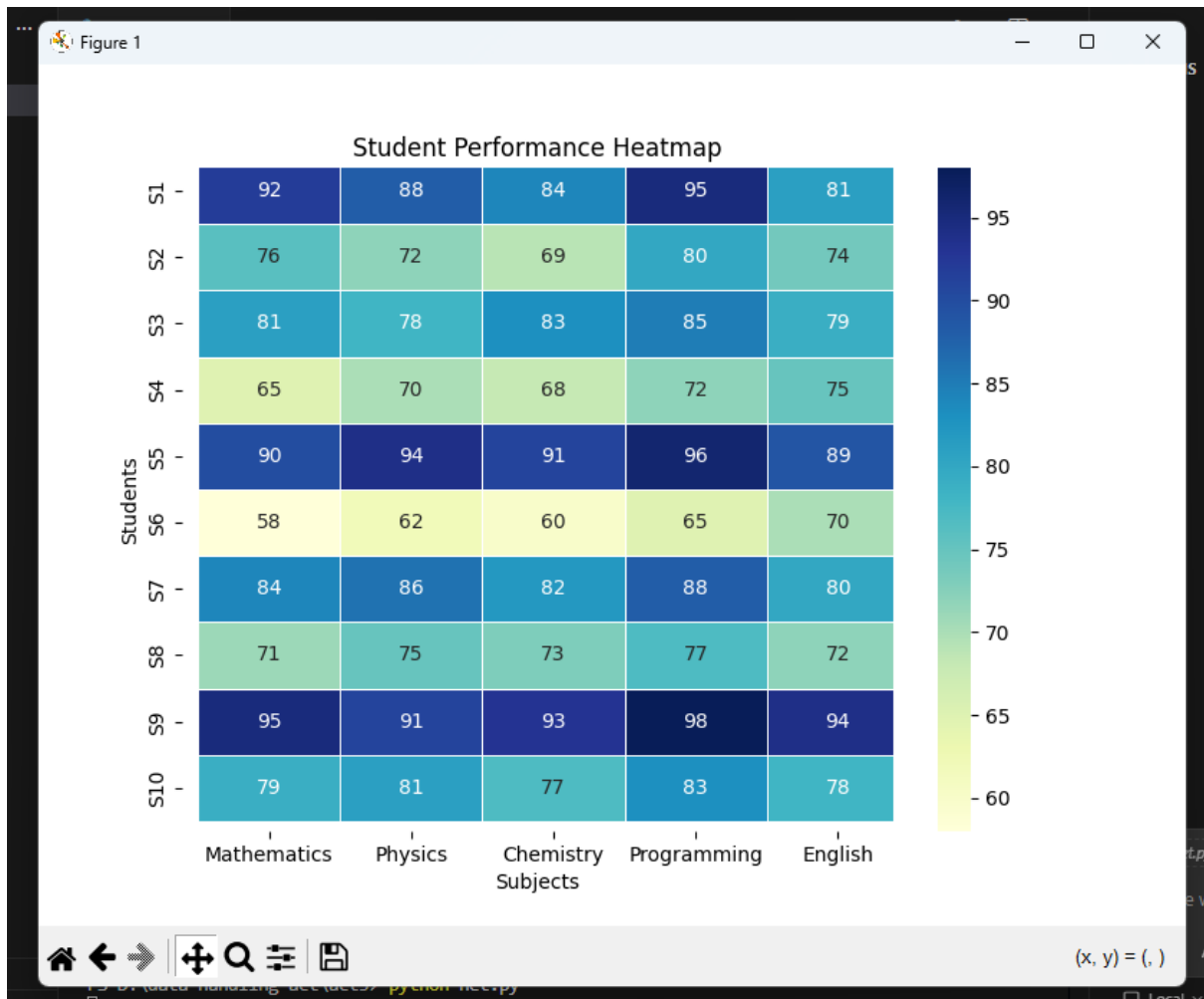


ACTIVITY 1



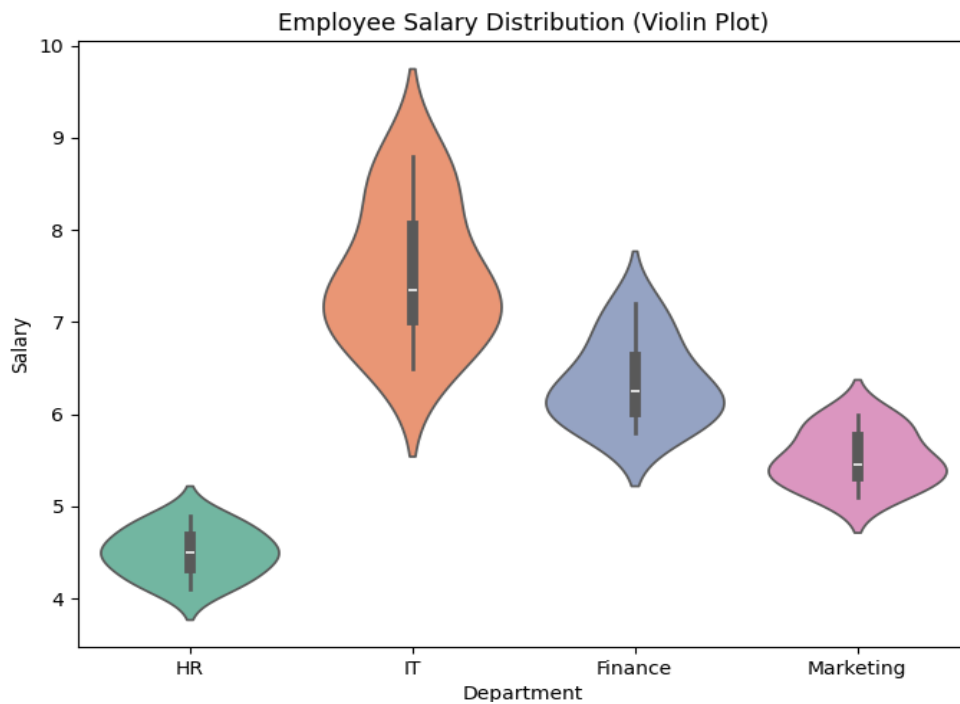
```
PS D:\data handling act\act3> python het.py
Highest Overall Performer:
S9 - 94.2

Highest Average Subject:
Programming - 83.9

Students Needing Support:
Average
S6 63.0
```

1. A heatmap provides a clear visual representation of data using colors.
2. It helps quickly identify high and low values without examining every number.
3. It makes it easy to compare the performance of different students and subjects.
4. It helps detect patterns, trends, and outliers in the dataset.
5. It is useful for summarizing large datasets in a simple and easy-to-understand format.

ACTIVITY 2



Standard Deviation (Variation):

```
HR      0.258199
IT      0.755425
Finance 0.455217
Marketing 0.301109
dtype: float64
```

Department with Largest Variation: IT

Median Salary:

```
HR      4.50
IT      7.35
Finance 6.25
Marketing 5.45
dtype: float64
```

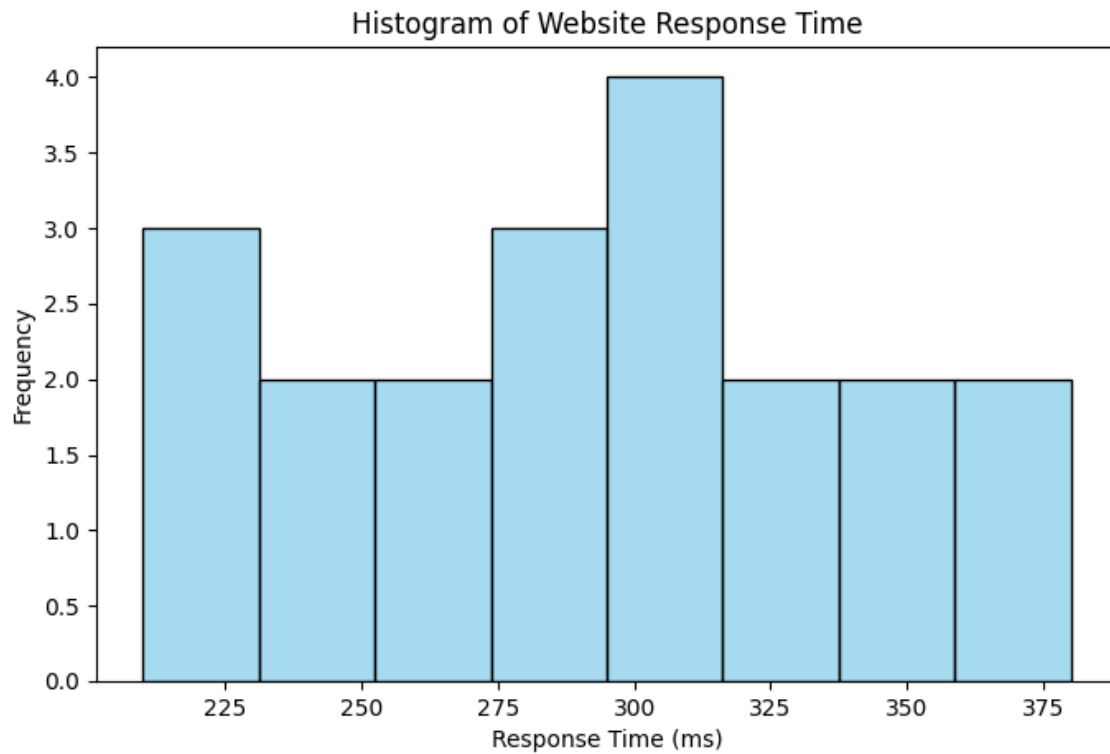
Department with Highest Median: IT

Best Visualization:

Violin plot better represents the distribution because it shows the data spread, density, and shape along with the distribution. Boxplot is useful for comparing medians, quartiles, and outliers.

1. Largest Variation: IT
2. Highest Median: IT
3. Best Plot: Violin Plot (shows the full distribution and density), while the Boxplot is better for summarizing median, quartiles, and outliers.

ACTIVITY 3



3. The response time data is slightly positively (right) skewed. Most observations are concentrated in the middle, with a few higher response times (up to 380 ms) extending the right tail.

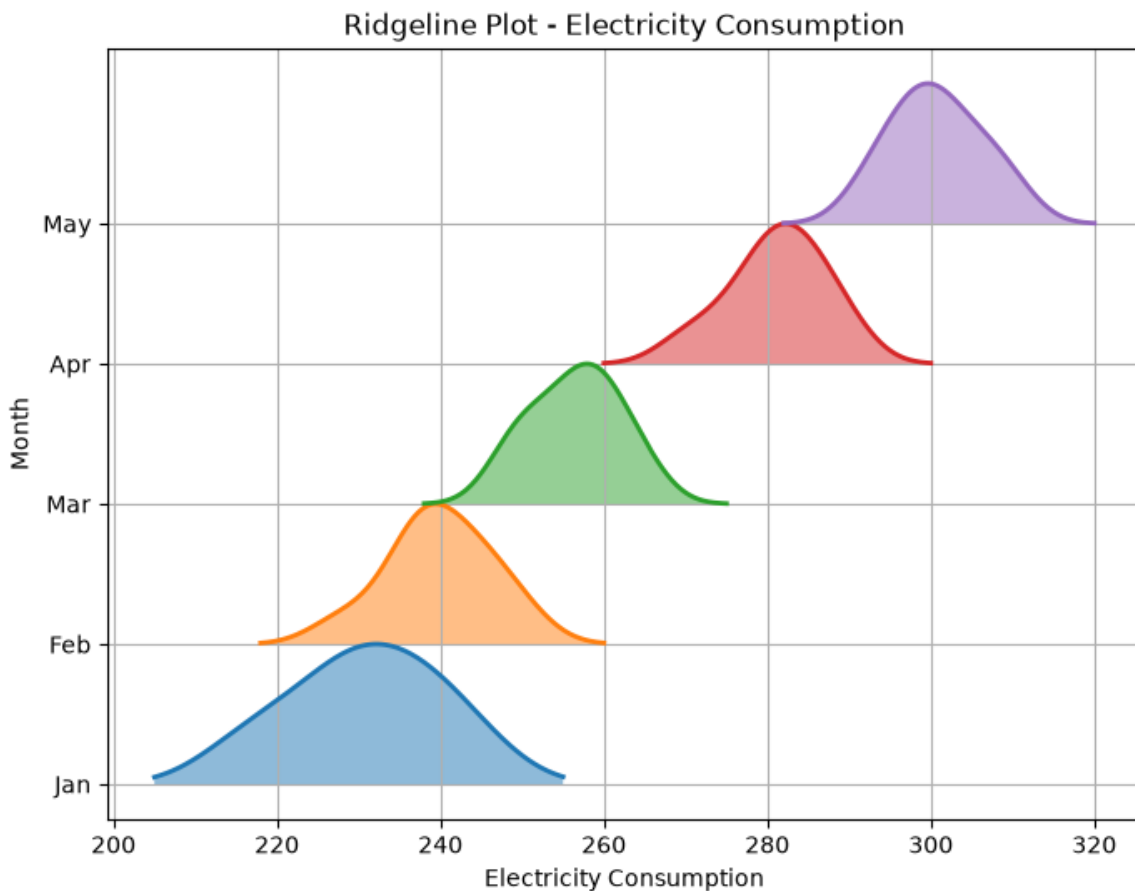
4. Common Range

Most website response times fall between 250 ms and 330 ms, indicating this is the most common response time range.

5. Which Plot Better Represents the Distribution?

The Density Plot better represents the distribution because it provides a smooth visualization of the data's overall shape and makes it easier to identify the distribution pattern. The Histogram is useful for showing the frequency of response times, but the density plot is more effective for understanding the distribution.

ACTIVITY 4



2. Highest Average Month: May (Highest average electricity consumption)

3. Widest Spread: May (Spread = 18 units, from 292 to 310)

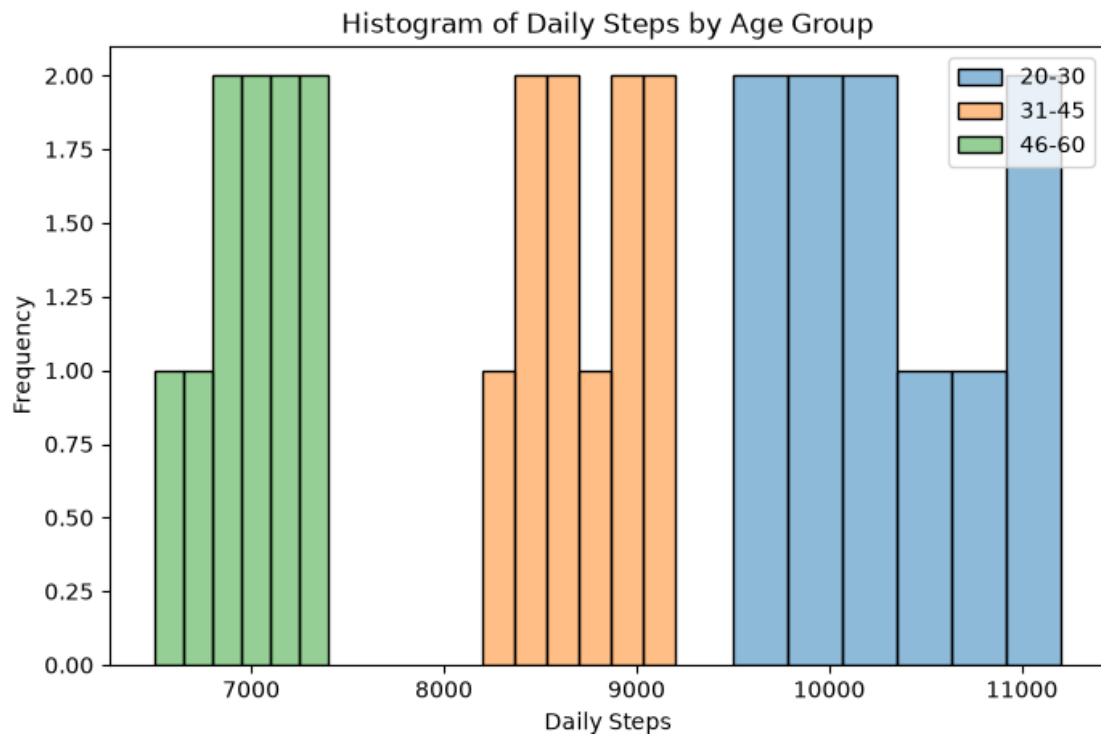
4. Describe Trend:

Electricity consumption increases steadily from January to May for all houses. The highest consumption occurs in May, showing a consistent upward trend over the months.

5. Why Ridgeline?

A ridgeline plot is useful because it displays the distribution of electricity consumption for each month in a single figure. It makes it easy to compare changes, observe trends over time, and identify differences in the spread and shape of the data across months.

ACTIVITY 5



4. Highest Median Group:
20–30 age group (highest median daily steps)

5. Best Visualization:
Boxplot is the best choice because it allows easy comparison of:

- Median daily steps
- Data spread (IQR)
- Variation between age groups
- Possible outliers

The density plot shows distribution shape, and histograms show frequency, but boxplots provide the clearest comparison between groups.